# Quantum Mekaniği Bölüm 2-3

## Tekrar

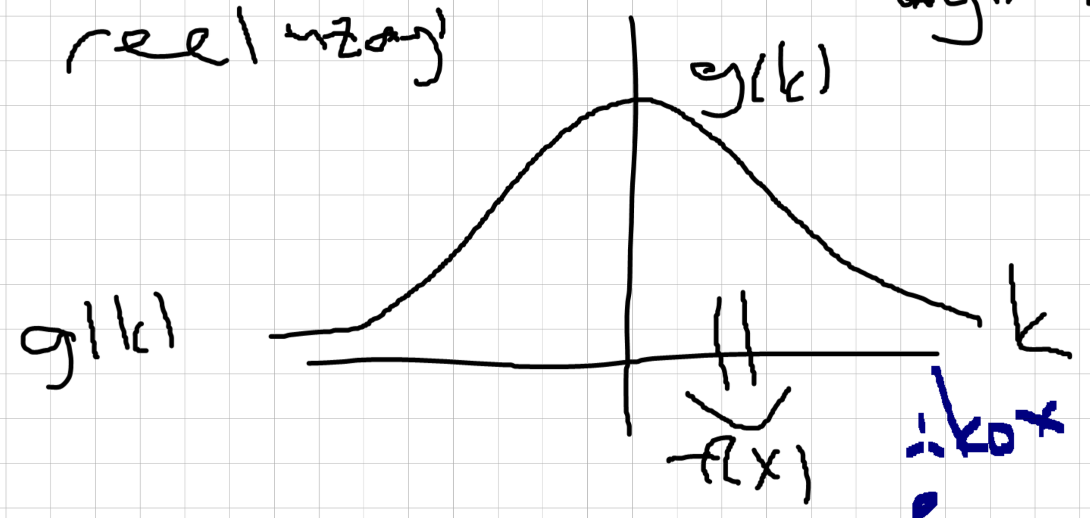
Bölüm 2

$$f(x) = \int_{-\infty}^{\infty} g(k) e^{ikx} dk$$

reel uzay — mom uzayı — ağırlık fonk.

$$\lambda = \frac{2\pi}{k} \Rightarrow \lambda k = 2\pi \quad (1/m)$$

$$g(k) = e^{-\alpha(k-k_0)^2}$$




$e^{ik_0x}$  $e^{ik_0x}$

$$(k-k_0) = k' \qquad dk = dk'$$

$$\Rightarrow f(x) = \int_{-\infty}^{+\infty} dk \cdot e^{-\alpha(k-k_0)^2} e^{i(k-k_0)x} \cdot e^{ik_0x}$$

$$\Rightarrow f(x) = e^{ik_0x}\int_{-\infty}^{+\infty} dk\, e^{-\alpha(k-k_0)^2} \cdot e^{-ik_0x} \cdot e^{ikx} \Rightarrow f(x) = e^{ik_0x}\int_{-\infty}^{+\infty} dk\, e^{-\alpha(k-k_0)^2} e^{i(k-k_0)x}$$

Gauss $\int e^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}}$ form lazım

$$\Rightarrow f(x) = e^{ik_0x}\int_{-\infty}^{+\infty} dk'\, e^{-\alpha k'^2} e^{ik'x} \Rightarrow e^{ik_0x}\int_{-\infty}^{+\infty} dk'\, e^{-\alpha k'^2 + ik'x}$$

$$\Rightarrow e^{ik_0x}\int dk'\, e^{-\alpha k'^2 + ik'x} = e^{ik_0x}\int dk' \cdot e^{\alpha\left(k' + \frac{ix}{2\alpha}\right)^2} \cdot e^{\frac{x^2}{2\alpha}}$$

$$\Rightarrow \alpha k'^2 + ik'x \Rightarrow (A+B)^2 + [C]$$

$$\alpha\left(\frac{k'^2}{\alpha} + i\frac{k'x}{\alpha}\right) \Rightarrow \alpha\left(\frac{k'}{} + \frac{ix}{2\alpha}\right)^2$$

$$\alpha\left(k'^2 - \frac{x^2}{4\alpha^2} + i\frac{k'x}{\alpha}\right) + [\ ]$$

$$[\ ] = \frac{+x}{2\alpha} \qquad \frac{x^2}{4\alpha^2}$$

$$e^{x^2/2\alpha} \cdot e^{ik_0x} \int dk''\, e^{\alpha k''^2} \Rightarrow \sqrt{\frac{\pi}{\alpha}}$$

$$k' + \frac{ix}{2\alpha} = k'' \quad dk' = dk'' \quad \text{dummy}$$

$$|g(k)|^2 = e^{-2\alpha(k-k_0)^2} \qquad |f(x)|^2 = \frac{\pi}{\alpha} e^{-x^2/2\alpha^2}$$

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$$w = w(k) \qquad w = ck \quad (c:\ \text{ışık hızı})$$

$$f(x,t) = \int_{-\infty}^{+\infty} dk\, g(k)\, e^{ikx - iw(k)t} \qquad g(k) = e^{-\alpha(k-k_0)^2}$$

$$f(x) = \sqrt{\frac{\pi}{\alpha}}\, e^{ik_0x}\, e^{-x^2/4\alpha^2}\ ?$$

$w(k)$ 'yı $k_0$ komşuluğunda yazalım

$$w(k) = w(k_0) + \frac{1}{1!}(k-k_0)\underbrace{\frac{dw}{dk}\Big|_{k_0}}_{V_g} + \frac{1}{2!}(k-k_0)^2\underbrace{\frac{d^2w}{dk^2}\Big|_{k_0}}_{B} + \cdots$$

$$w(k) = w(k_0) + (k-k_0)V_g + (k-k_0)^2 B$$

$$\Rightarrow f(x,t) = \int_{-\infty}^{+\infty} dk\, g(k)\, e^{ikx - iw(k)t} \Rightarrow \int dk\, e^{-\alpha(k-k_0)^2} \cdot e^{ikx - i[w(k_0) + (k-k_0)V_g + (k-k_0)^2 B]t}$$

$$\Rightarrow \int dk\, e^{-\alpha(k-k_0)^2} \cdot e^{ikx} \cdot e^{-ik_0x} \cdot e^{-iw(k_0)t} \cdot e^{-i(k-k_0)V_g t} \cdot e^{-i(k-k_0)^2 Bt}$$

$$e^{-i\omega(k)t \underbrace{-ik_0 x}_{?}} \int dk\, \underbrace{e^{-\alpha(k-k_0)^2 - i(k-k_0)x - i(k-k_0)\sqrt{g}t - i(k-k_0)^2 Bt}}_{\#}$$

$(k-k_0) = k'$

$dk = dk'$

$$\# \Rightarrow \int dk'\, e^{-\alpha k'^2 - ik'x - ik'\sqrt{g}t - ik'^2 Bt}$$

$$\Rightarrow \int dk'\, e^{-(\alpha k'^2 + ik'^2 Bt + ik'x + ik'\sqrt{g}t)}$$

$$\Rightarrow \int dk'\, e^{-(k'^2(\alpha + iBt) + ik'(x+\sqrt{g}t))}$$

$\alpha + iBt = A$

$i(x+\sqrt{g}t) = B$

$$\int dk'\, e^{-(k'^2 A + k'B)}$$

$$\Rightarrow -Ak'^2 - Bk' \Rightarrow -A\left(k' + \frac{B}{2A}\right)^2 + [\ ]$$

$$-A\left(k'^2 + \frac{B}{A} + \frac{B^2}{4A^2}\right) + \left[\underline{-\frac{B^2}{4A^2}}\right]$$

$$\left(\frac{B}{2A}\right)^2$$

$$\Rightarrow \int dk'\, e^{-A\left(k'+\frac{B}{2A}\right)^2} \cdot e^{+\left(\frac{B}{2A}\right)^2} \Rightarrow e^{\left(\frac{B}{2A}\right)^2} \int dk'\, e^{-A\left(k'+\frac{B}{2A}\right)^2}$$

$k' + \frac{B}{2A} = k'' \Rightarrow$

$dk' = dk''$

$$\Rightarrow e^{A\left(\frac{B}{2A}\right)^2} \int dk''\, e^{-Ak''^2} \Rightarrow e^{\left(\frac{B}{2A}\right)^2} \sqrt{\frac{\pi}{A}}$$

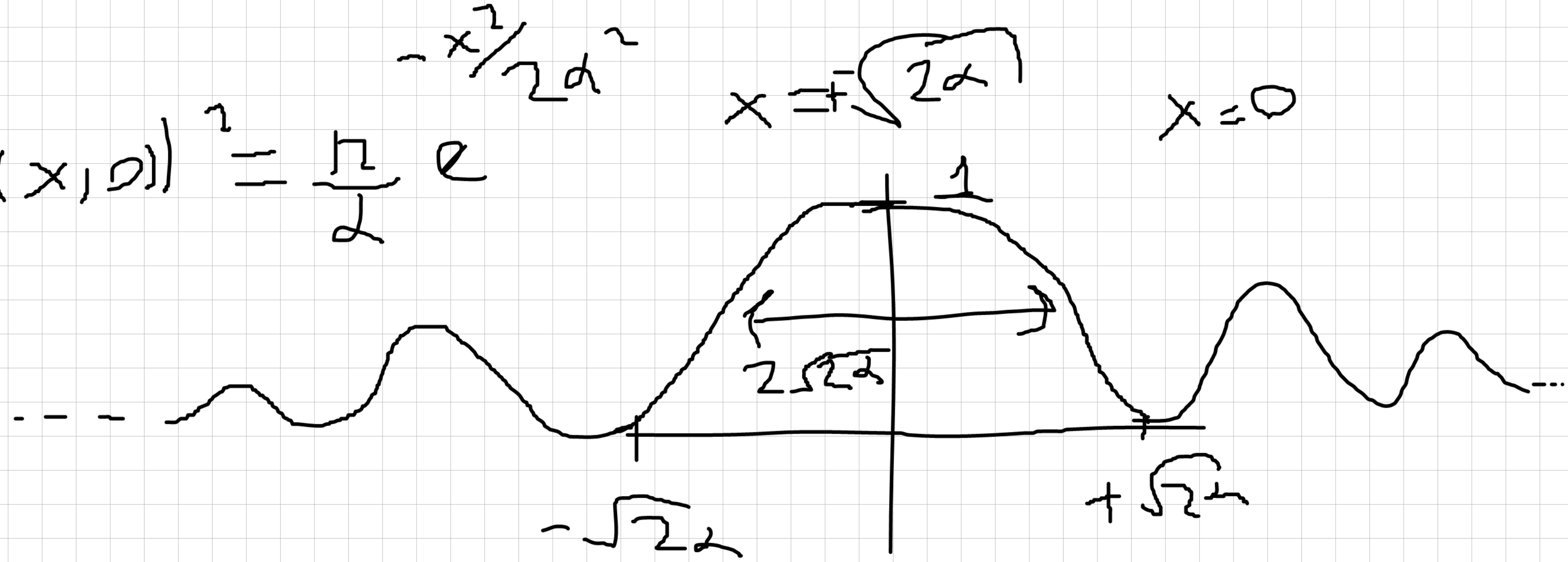
$$\boxed{B = i(x+\sqrt{g}t) \qquad A = \alpha + iBt}$$

$$e^{\left(-i\omega(k)t - ik_0 x + A\left(\frac{B}{2A}\right)^2\right)} \sqrt{\frac{\pi}{A}}$$

$$\Rightarrow \sqrt{\frac{\pi}{\alpha + iBt}}\, e^{-i\omega(k)t - ik_0 x - (x+\sqrt{g}t)^2/4(\alpha+iBt)} = f(x,t)$$

$$|f(x,t)|^2 = \frac{\pi}{\sqrt{\alpha^2 + B^2t^2}}\, e^{-(x-\sqrt{g}t)^2/2(\alpha^2+B^2t^2)}$$

$$t = 0 \Rightarrow \frac{\pi}{\alpha} e^{-x^2/2\alpha^2} \Rightarrow |f(x,0)|^2 = \frac{\pi}{\alpha} e^{-x^2/2\alpha^2}$$



$$f(x,t) = \int_{-\infty}^{+\infty} dk\, g(k)\, e^{ikx - i\omega(k)t} \qquad E = \frac{p^2}{2m} \qquad \boxed{p = \hbar k} \Rightarrow \boxed{E = \frac{p^2}{2m} = \hbar\omega}$$

$$dp = \hbar\, dk \Rightarrow \boxed{\frac{dp}{\hbar} = dk} \qquad \boxed{E = \hbar\omega}$$

$$\Rightarrow \underbrace{f(x,t)}_{\psi} = \int_{-\infty}^{+\infty} \frac{dp}{\underbrace{\hbar\, g(p)}_{\phi}}\, e^{i\frac{p}{\hbar}x - i\frac{E}{\hbar}t}$$

$$\Rightarrow \Psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} dp\, \phi(p)\, e^{\frac{i}{\hbar}(px - \overset{p^2/2m}{E}t)} \qquad \frac{1}{\sqrt{2\pi\hbar}} \int dp\, \phi(p)\, e^{i/\hbar(px - Et)} = \partial$$

$$\Rightarrow \frac{d\psi}{dx}, \frac{d\psi}{dt} \Rightarrow \frac{d\psi}{dx} = \frac{i}{\hbar} p\, \partial \qquad \underbrace{\frac{d^2\psi}{dx^2} = -\frac{1}{\hbar^2} p^2 \partial \qquad \frac{d\psi}{dt} = \frac{-i}{\hbar} \overset{E}{\frac{p^2}{2m}} \partial}$$

$$\frac{d^2\psi}{dx^2} = \frac{-1}{\hbar^2} p^2 \partial \qquad \frac{d\psi}{dt} = \frac{-i}{\hbar} \frac{p^2}{2m} \partial$$

$$\Rightarrow [\quad] \frac{d^2\psi}{dx^2} = [\quad] \frac{d\psi}{dt} \Rightarrow -\frac{1}{\hbar^2} \overset{x}{[\quad]} = \frac{-i}{2m\hbar} \overset{t}{[\quad]}$$

$$\Rightarrow -\frac{1}{\hbar^2}\left[\frac{1}{2m}\right] = \frac{-i}{2m\hbar}[\quad]$$

$$\Rightarrow -1\left[\frac{1}{2m} i\right] = \frac{-i}{2m\hbar}\left[\frac{1}{\hbar}\right]$$

$$\Rightarrow \left\{ \frac{i}{2m} \frac{d^2\psi}{dx^2} = \frac{1}{\hbar} \frac{d\psi}{dt} \right\} \times \hbar^2$$

$$i \cdot i = -1$$

$$\Rightarrow \left\{ \frac{i\hbar^2}{2m} \frac{d^2\psi}{dx^2} = \hbar \frac{d\psi}{dt} \right\} \times (i)$$

$$\Rightarrow \boxed{-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = i\hbar \frac{d\psi}{dt}}$$

Schrodinger Denklemi